What is Deep Learning

Deep learning (DL) is a specialized subfield of machine learning (ML) that utilizes multilayered artificial neural networks to perform complex tasks such as classification, regression.

How It is Different from Traditional ML

It is distinguished from traditional machine learning by its ability to learn hierarchies of concepts, where each concept is defined by its relation to simpler ones

The Meaning of "Deep"

The adjective "deep" refers to the number of layers through which data is transformed within the network.

How many Layers Then

Modern deep learning models can range from three layers to several hundred or even thousands.

Any Specialty?

Automatic Feature Learning

A defining characteristic of deep learning is its ability to extract useful features from raw data automatically.
Because of its many layers, the model analyzes raw data directly and discovers for itself which patterns matter most for the task

For example, in an image recognition model, early layers might detect simple edges, middle layers might detect shapes, and final layers might recognize entire objects like faces.

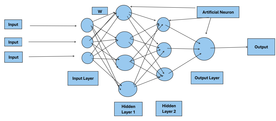
Traditional Machine Learning: Often requires "feature engineering," where humans must manually decide what information (features) the model should focus on.

Architecture and Biological Inspiration

Deep learning architectures are loosely inspired by the biological neural networks of animal brains, though they are now guided by mathematical and engineering principles rather than being strict simulations of biological function.

Components

These networks consist of connected units called artificial neurons, organized into an input layer, hidden layers, and an output layer.



WX+B

W-Weight
X-Input
B-bias

Mathematical Function

A deep network can be viewed as a massive mathematical function that maps input values to output values by composing many simpler functions.

Model Training

Models are "trained" using optimization algorithms—primarily gradient descent—and the backpropagation algorithm to adjust the internal "weights" of the connections until the model performs accurately.

Key Requirements

Data Requirements

DL models typically require large datasets (often millions of examples) to reach their full potential and avoid simply memorizing training data.

Computing Power

Training these complex models requires significant resources, typically involving powerful GPUs (graphics processing units) or cloud computing services.

Practical Applications

Computer Vision

Identifying objects in photos, facial recognition, and medical scan analysis

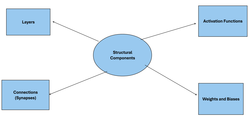
Natural Language Processing (NLP)

Powering machine translation (e.g., Google Translate), voice assistants (e.g., Alexa, Siri), and sentiment analysis.

Artificial Neural Network

Artificial Neural Networks (ANNs), also known as connectionist systems, are computing systems inspired by the biological neural networks that constitute animal brains.

These systems are designed to learn and progressively improve their ability to perform tasks—such as image recognition or speech processing—by considering examples, generally without being programmed with task-specific rules.



Layers

Neurons are arranged in an input layer (which receives data), one or more hidden layers (where transformations occur), and an output layer (which provides the final result).

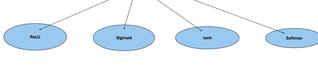
Connections (Synapses)

Each connection between neurons can transmit a signal. The receiving neuron processes the signal and then signals downstream neurons.

Weights and Biases

Each connection has a weight that increases or decreases the strength of the signal it sends. Biases are additional parameters used to shift the activation function's output, helping the network better fit the data.

Activation Functions

These mathematical functions are applied to the input of a neuron to determine its output activation, often introducing nonlinearity to the model.

Types Of funtion